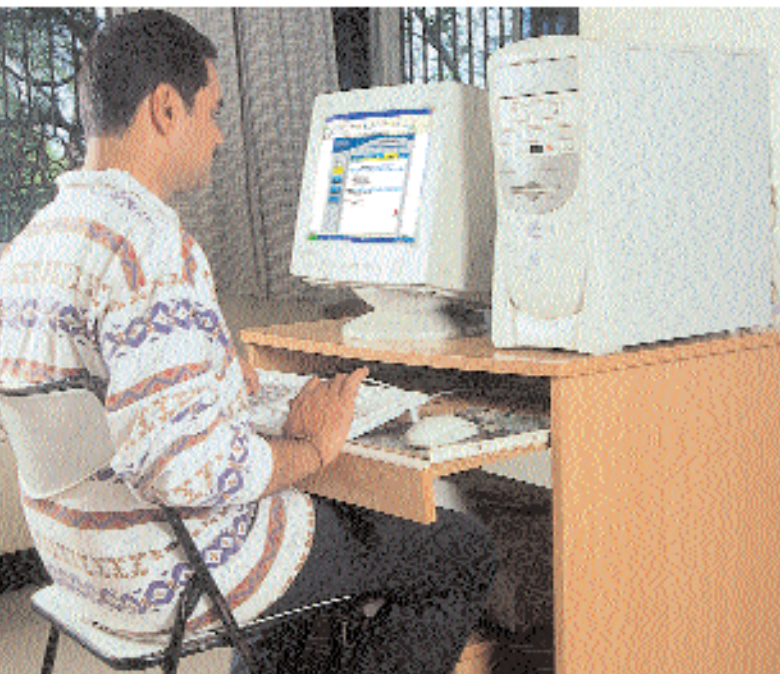




Wires are a tangly web that can get on anyone's nerves. Wireless Networking solves the problem, but like a BMW, costs more for the comfort it offers

(Wire)Less is More

Computers can be used to talk to each other by connecting them wirelessly, using the latest Wi-Fi access points and adapters. Connecting machines situated on different floors, or in rooms several metres apart, will require long cables and a few hundred holes drilled all over the place. With wireless connectivity no drilling of holes or investment in cables is required—just plug the hardware into the PC and connect.

We decided to experiment with wireless connectivity, and that too with the cheapest ingredients—D-Link's DW-900AP+ access point, a DWL-650+ PCMCIA card for the laptop and a DWL-520+ PCI-based card for our desktop PC. These components are also capable of securing home users sufficiently.

We considered a typical office environment, where a wired LAN exists but a few people want to have the freedom and portability that wireless offers. Thus, a hybrid LAN is formed, with some computers connecting with normal ethernet cards, and a few connecting with wireless

adapters. In order to get seamless connectivity within the hybrid LAN, an interface between the wired and wireless LAN is required. This is where the Access Point comes in.

In order to achieve this real world scenario in our test centre, we connected two desktop PCs, one called 'master', the other called 'client', and a laptop. The access point and PCI and PCMCIA interface adapters will take care of the Wi-Fi. The master PC had a built-in LAN card, and a modem provided Internet access. We used Internet connection sharing software (WinGate or WinProxy5), to enable the client to access the Internet via the master.

In all, we needed an access point and wireless adapters. If we consider the current price of the D-Link Access Point, and PCI and PCMCIA-based adapters, then the total solution cost us approximately Rs 18,400. This amount appears pretty high as compared to a wired LAN, but is a boon for small businesses which operate from rented offices—especially if they are not allowed to alter the interior in

any way, such as drilling holes. Not only is it convenient to set up, it's also very simple to move to a different location.

If you're looking to connect five or ten PCs on the same floor, and are willing to go through the hassles of networking them, then wired LAN is the cheaper option. But, if you want to have the latest technology, with a system that most 15-year olds could set up, and also have an aesthetically pleasant look to your office, then wireless is the way to go.

Set up and configuration

Setting up a Wireless LAN (WLAN) is really easy. However, depending on your network type, the requirements change. There are two ways to go about this, either you have a purely wireless LAN, where every computer is freely portable, or you share a LAN with wired computers, in a hybrid of wired and wireless LAN.

To configure a purely wireless network, you need to install the cards and configure them to run in Ad-Hoc mode, which can be done from the network connection wizard in Windows XP or

the utility bundled with the Adapter card. Now, install a proxy server on the computer with the Internet connection (master) and set the others to access the Net through it. That's it; your network is up and running.

However, a word of caution; when buying wireless hardware, make sure they are compatible—802.11b (Wi-Fi) will not work with 802.11a/g hardware. Also, wireless equipment from different vendors may or may not work together. Buy